Math 492

An Introduction to Rigid Analytic Geometry

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Outline

- Varieties
- Schemes
- Rigid Analytic Varieties

The Classical Correspondence

There is a bijective correspondence between geometry and algebra:

$$\Big\{\text{affine varieties}\Big\} \quad \longleftrightarrow \quad \Big\{\text{prime ideals}\Big\}$$

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Definition

For $S \subset k[x_1, \ldots, x_n]$, the zero set $Z(S)$ of S is the set

$$Z(S) = \{p \in \mathbb{A}^n \mid f(p) = 0 \text{ for all } f \in S\}.$$

A subset $X \subset \mathbb{A}^n$ is called an algebraic set if $X = Z(S)$ for some $S \subset k[x_1, \ldots, x_n]$. Similarly, the ideal of $X \subset \mathbb{A}^n$ is the set

$$I(X) = \{f \in k[x_1, \ldots, x_n] \mid f(p) = 0 \text{ for all } p \in X\}.$$

Note that Z and I are inclusion-reversing.

Example

The zero set of $0 \subset k[x_1, ..., x_n]$ is $\mathbb{A}^n$: $Z(0) = \mathbb{A}^n$.
The ideal of $\mathbb{A}^n$ is 0 : $I(\mathbb{A}^n) = 0$.

Moreover, we have

$$Z(I(X)) = X$$

for any $X \in \mathbb{A}^n$.

Note that Z and I are inclusion-reversing.

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The zero set of $0 \subset k[x_1, ..., x_n]$ is $\mathbb{A}^n$: $Z(0) = \mathbb{A}^n$.
The ideal of $\mathbb{A}^n$ is 0 : $I(\mathbb{A}^n) = 0$.

Moreover, we have

$$Z(I(X)) = X$$

for any $X \in \mathbb{A}^n$. However,

$$I(Z(S)) \neq S$$

since

$$I(Z(x^2)) = (x) \neq (x^2)$$

in $k[x]$.

Definition

Given an ideal $J \subset k[x_1, \ldots, x_n]$, the radical of J is the set

$$\sqrt{J} = \{f \in k[x_1, \ldots, x_n] \mid f^m \in J \text{ for some } m > 0\}.$$

The ideal J is said to be radical if $J = \sqrt{J}$.

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Proposition

The maps

$$\{\text{algebraic sets in } \mathbb{A}^n\} \underset{Z}{\overset{I}{\rightleftarrows}} \{\text{radical ideals of } k[x_1, \ldots, x_n]\}$$

are bijections that are inverses of each other.

Proof

By Nullstellensatz, $I(Z(J)) = \sqrt{J}$. Since J is radical, $\sqrt{J} = J$.

Hilbert's Nullstellensatz

Let $J \subset k[x_1, \dots, x_n]$ be an ideal. Suppose that $f \in k[x_1, \dots, x_n]$ vanishes on $Z(J)$. Then $f^m \in J$ for some $m > 0$.

Definition (Affine Variety)

An algebraic set $X \subset \mathbb{A}^n$ is called an affine variety if it cannot be written as $X = X_1 \cup X_2$, where $X_1, X_2 \subset \mathbb{A}^n$ are algebraic sets with $X_1, X_2 \subsetneq X$.

Reducible

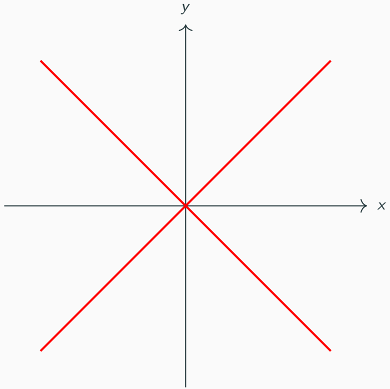
$X = Z(y-x) \cup Z(y+x) = Z(y^2 - x^2)$

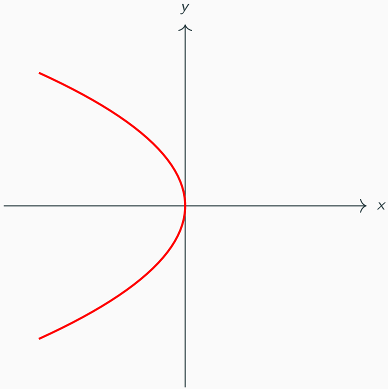
$I(X) = ((y-x)(y+x))$

Irreducible

$X = Z(x + y^2)$

$I(X) = (x + y^2)$





Proposition

Let $X \subset \mathbb{A}^n$ be an algebraic set. Then X is an affine variety if and only if $I(X)$ is a prime ideal.

Observe that the maps I and Z induce a one-to-one correspondence

$$\{\text{affine varieties in } \mathbb{A}^n\} \underset{Z}{\overset{I}{\rightleftarrows}} \{\text{prime ideals in } k[x_1, \ldots, x_n]\}.$$

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So far, we have

$$\begin{tikzcd}
\left\{\text{affine varieties in } \mathbb{A}^n\right\} \arrow[r, leftrightarrow] \arrow[d, hook] & \left\{\text{prime ideals in } k[x_1, \ldots, x_n]\right\} \arrow[d, hook] \\
\left\{\text{algebraic sets in } \mathbb{A}^n\right\} \arrow[r, leftrightarrow] & \left\{\text{radical ideals in } k[x_1, \ldots, x_n]\right\}.
\end{tikzcd}$$

Definition

Let $X \subset \mathbb{A}^n$ be an affine variety. The coordinate ring $k[X]$ of X is the ring $k[X] = k[x_1, \ldots, x_n]/I(X)$.

By definition, we have

$$I(X) \text{ is a prime ideal} \iff k[X] \text{ is an integral domain.}$$

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$$I(X) \text{ is a prime ideal} \iff k[X] \text{ is an integral domain.}$$

Proposition

Let A be a k-algebra. Then $A = k[X]$ for some affine variety X iff A is an integral domain and finitely generated as a k-algebra.

In the case of the above proposition, we call A an affine k-domain. We obtain another correspondence

$$\Big\{\text{affine varieties}\Big\} \longleftrightarrow \Big\{\text{affine } k\text{-domains}\Big\}.$$

Transitioning to Schemes

Replacing affine k-domains with an arbitrary commutative ring with identity, we wish to establish a similar correspondence:

$$\begin{tikzcd}
\left\{\text{affine varieties}\right\} \arrow[r, leftrightarrow] \arrow[d] & \left\{\text{affine } k\text{-domains}\right\} \arrow[d] \\
\left\{\phantom{\text{affine varieties}}\right\} \arrow[r, leftrightarrow] & \left\{\text{rings}\right\}.
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\left\{\text{affine varieties}\right\} \arrow[r, leftrightarrow] \arrow[d] & \left\{\text{affine } k\text{-domains}\right\} \arrow[d] \\
\left\{\text{affine schemes}\right\} \arrow[r, leftrightarrow] & \left\{\text{rings}\right\}.
\end{tikzcd}$$

Spectrum of a Ring

Let $X \subset \mathbb{A}^n$ be an affine variety. By the Nullstellensatz, one has

$$\Big\{\text{points } p = (a_1, \ldots, a_n) \in X\Big\} \quad \longleftrightarrow \quad \Big\{\text{max ideals } \mathfrak{m}_p = (x_1 - a_1, \ldots, x_n - a_n)\Big\}.$$

The evaluation homomorphism

$$\mathrm{ev}_p : k[x_1, \ldots, x_n] \to k, \;\; f \mapsto f(p)$$

has $\ker(\mathrm{ev}_p) = \mathfrak{m}_p$. Then $f \in \mathfrak{m}_p$ for all $f \in I(X)$.

By the correspondence theorem,

$$\left\{\text{points of } X\right\} \quad \longleftrightarrow \quad \left\{\text{max ideals of } k[x_1, \ldots, x_n] \supset I(X)\right\}.$$

If $\mathfrak{p} \supset I(X)$ is a prime ideal that is not maximal, it corresponds to an affine variety $Y = Z(\mathfrak{p}) \subset Z(I(X)) = X$. Then

$$\left\{\text{subvarieties of } X\right\} \quad \longleftrightarrow \quad \left\{\text{prime ideals of } k[x_1, \ldots, x_n] \supset I(X)\right\}.$$

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Combining,

$$X \sim \text{prime ideals of } k[X].$$

Spec R

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Similar to varieties, let $V(I) = \{\mathfrak{p} \in \mathrm{Spec}(R) \,|\, I \subset \mathfrak{p}\}$ for an ideal I in R.

Proposition

Let R be a ring. We have the following properties:

(a) For any pair of ideals I, J of R, we have $V(I) \cup V(J) = V(I \cap J)$.

(b) Let $(I_\lambda)_\lambda$ be a family of ideals of R. Then $\bigcap_\lambda V(I_\lambda) = V(\sum_\lambda I_\lambda)$.

(c) $V(R) = \emptyset$ and $V(0) = \mathrm{Spec}(R)$.

Zariski Topology

By the previous proposition, Spec R can be given a topology with $V(I)'$s as closed sets. This topology is called the Zariski topology.

The sets of the form

$$D(f) = \text{Spec } R \setminus V(f) = \{\mathfrak{p} \in \text{Spec } R : f \notin \mathfrak{p}\}$$

form a basis for the Zariski topology since

$$U = \operatorname{Spec} R \setminus V(I) = \operatorname{Spec} R \setminus \bigcap_{f \in I} V(f) = \bigcup_{f \in I} (\operatorname{Spec} R \setminus V(f)) = \bigcup_{f \in I} D(f),$$

where $I = \sum_{f \in I} (f)$.

Definition

Let X be a topological space. A presheaf $\mathcal{F}$ of abelian groups consists of

i) an abelian group $\mathcal{F}(U)$ for every open subset U of X, and

ii) a group homomorphism $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$ for every pair of open subsets $V \subset U$

which satisfy the following conditions:

1) $\mathcal{F}(\emptyset) = 0$;

2) $\rho_{UU} = id_{\mathcal{F}(U)}$;

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Example

Let $C(U, \mathbb{R})$ be the ring of continuous real-valued functions on U. Then C is a presheaf.

Let $\{U_i\}_i$ be an open covering of U and let $s \in \mathcal{F}(U)$. We say $\mathcal{F}$ is a sheaf if we have the following additional properties:

4) If $s|_{U_i} = 0$ for every i, then $s = 0$.

5) If $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for each pair i, j. Then there exists $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for every i.

Example

C is also a sheaf.

Let $U \subset \mathbb{R}^m$ be open and let $F : U \to \mathbb{R}^n$ be a continuous function. Then F is $\mathcal{C}^\infty$ if and only if $f \circ F : U \to \mathbb{R}$ is $\mathcal{C}^\infty$ for each $\mathcal{C}^\infty$ map $f : \mathbb{R}^n \to \mathbb{R}$.

Locally Ringed Spaces

Let $U \subset \mathbb{R}^m$ be open and let $F : U \to \mathbb{R}^n$ be a continuous function. Then F is $\mathcal{C}^\infty$ if and only if $f \circ F : U \to \mathbb{R}$ is $\mathcal{C}^\infty$ for each $\mathcal{C}^\infty$ map $f : \mathbb{R}^n \to \mathbb{R}$.

Observations

(i) The set

$$\mathcal{C}^\infty(U) = \{F : U \to \mathbb{R}^n \,|\, F \text{ is } \mathcal{C}^\infty\}$$

is a ring.

(ii) Either $F(x) = 0$ or $1/F$ exists in some neighborhood of x. In other words, the stalk at x is a local ring with the maximal ideal consisting of F with $F(x) = 0$.

Definition (Locally Ringed Spaces)

Let X be a topological space with a sheaf of rings $\mathcal{O}_X$ on it. The pair $(X, \mathcal{O}_X)$ is called a locally ringed space if $\mathcal{O}_{X,x}$ is a local ring for all $x \in X$.

We denote by $\mathfrak{m}_x$ the maximal ideal of $\mathcal{O}_{X,x}$ and by $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ its quotient field.

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Example

A locally ringed space $(M, \mathcal{O}_M)$ is called a smooth manifold if there exists an open covering $M = \bigcup_{i \in I} U_i$ such that for all $i \in I$, there exist an open set $V \subset \mathbb{R}^n$ and an isomorphism $(U_i, \mathcal{O}_M|_{U_i}) \cong (V, \mathcal{C}^\infty(V))$.

The Structure Sheaf

Let R be a ring, and let $f \in R$. Define the value of f at $\mathfrak{p} \in \operatorname{Spec} R$ to be the image of f under

$$R \longrightarrow R/\mathfrak{p} \hookrightarrow \kappa(\mathfrak{p}).$$

Then f can be viewed as a map

$$f : \operatorname{Spec} R \longrightarrow \coprod_{\mathfrak{p} \in \operatorname{Spec} R} \kappa(\mathfrak{p}).$$

For the functions on Spec R, we want

$$\mathcal{O}_{\text{Spec } R}(\text{Spec } R) = R.$$

For basis elements $D(f)$, we know f is non-vanishing. So we set

$$\mathcal{O}_{\text{Spec } R}(D(f)) := R_f = \left\{ \frac{r}{f^n} : r \in R, n \geq 0 \right\}.$$

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The pair $(\text{Spec } R, \mathcal{O}_{\text{Spec } R})$ is called an affine scheme.

Definition

A locally ringed space $(X, \mathcal{O}_X)$ is called a scheme if there exists an open covering $\{U_i\}_{i \in I}$ of X such that $(U_i, \mathcal{O}_X|_{U_i})$ is isomorphic to an affine scheme.

Non-Archimedean Valued Fields

Definition

A non-Archimedean valued field is a field k equipped with a function $|| : k \to \mathbb{R}_{\geq 0}$ such that

(i) $|x| = 0 \iff x = 0$,

(ii) $|xy| = |x|\,|y|$ for all $x, y \in k$, and

(iii) $|x + y| \leq \max(|x|, |y|)$.

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Example (p-adic Valuations)

Fix a prime number p. Any non-zero rational number $x \in \mathbb{Q}$ can be written as $x = p^k \frac{a}{b}$, where $k \in \mathbb{Z}$ and p divides neither a nor b. We define the p-adic absolute value $||_p : \mathbb{Q} \to \mathbb{R}_{\geq 0}$ by

$$|x|_p = p^{-k} \quad \text{and} \quad |0|_p = 0.$$

Geometric Consequences

The metric $d(x, y) = |x - y|$ has the following properties:

- A series $\sum_{\nu=0}^{\infty} a_\nu$ converges $\iff a_\nu \to 0$.
- If $|x| \neq |y|$, then $|x + y| = \max(|x|, |y|)$.
- Every triangle is isosceles.
- Any point inside an open ball is its center.
- Open balls are closed.
- The topology induced is totally disconnected, i.e., the only connected components are singletons.

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- The topology induced is totally disconnected, i.e., the only connected components are singletons.

Because the space is totally disconnected, we can define locally constant functions that are 1 on one ball and 0 on another. Consequently, analytic continuation fails.

Tate Algebras

Definition

Let k be complete. The Tate algebra $T_n(k)$ is the ring of convergent power series

$$T_n(k) = \left\{ \sum_{\nu \in \mathbb{N}^n} a_\nu X^\nu \;\middle|\; a_\nu \in k, \lim_{|\nu| \to \infty} |a_\nu| = 0 \right\}.$$

Let

$$\operatorname{Max} T_n = \{x \subset T_n : x \text{ is a maximal ideal of } T_n\}$$

and

$$B^n(k) = \{(x_1, \ldots, x_n) \in k^n : \max |x_i| \le 1\}.$$

Maximal Ideals and Points

Let $x = (x_1, \ldots, x_n) \in B^n(\overline{k})$, and set

$$\mathfrak{m}_x = \{f \in T_n : f(x) = 0\}.$$

Then

$$h_x : T_n \to k(x_1, \ldots, x_n),\ \ f \mapsto f(x)$$

satisfies $\ker h_x = \mathfrak{m}_x$ and $T_n / \ker h_x \cong k(x_1, \ldots, x_n)$. So $\mathfrak{m}_x \in \operatorname{Max}\ T_n$.

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Conversely, given $\mathfrak{m} \in \mathrm{Max}\ T_n$, the value of f at $\mathfrak{m}$ is the image under the composition

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Conversely, given $\mathfrak{m} \in \mathrm{Max}\ T_n$, the value of f at $\mathfrak{m}$ is the image under the composition

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Theorem

The mapping $x \mapsto \mathfrak{m}_x$ is surjective and has finite fibers.

Affinoid Algebras

A k-algebra A is called an affinoid algebra if there is a surjective k-algebra homomorphism $\pi : T_n(k) \to A$.

Just as we defined affine k-domains as quotients of $k[x_1, \ldots, x_n]$, we define affinoid algebras as quotients of $T_n(k)$.

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Affinoid Varieties

A k-affinoid variety is a pair Sp $A = (\text{Max A}, A)$ where A is an affinoid algebra.

Similarly, affinoid varieties are analogs of affine schemes.

Finally, we can define global varieties:

Rigid Analytic Varieties

A rigid analytic variety over k, or a k-analytic variety, is a locally G-ringed space such that X admits an admissible open covering $\{X_i\}_{i \in I}$ such that $(X_i, \mathcal{O}_{X_i})$ is isomorphic to a k-affinoid variety for each $i \in I$.

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